MIL-R-25085C(USAF)
3 June 1959

Superseding
MIL-R-25085B(USAF)
1 November 1957

MILITARY SPECIFICATION

RACK, BOMB, AIRCRAFT, ELECTRIC, TYPE MB-3A

1. SCOPE

1.1 This specification establishes the requirements for one type of 30 inch lug span, 24V dc bomb rack, designated Type MB-3A and designated for use with stores weighing up to and including 5000 pounds.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiber
PPP-T-60	Tape; Pressure Sensitive Adhesive, Waterproof-For Packaging and Sealing

Military

MIL-P-116	Preservation, Methods Of
MIL-B-138	Boxes, Wood, Fiberboard-Lined For Overseas Shipment (For Weight Contents Not Exceeding 500 Pounds)
MIL-G-3278	Grease; Aircraft and Instruments (For Low and High Temperatures)
MIL-D-5028	Drawings and Data Lists: Preparation Of Manufacturers' (For Production Aeronautical and Associated Equipment)
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification For

FSC 1095

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MIL-W-6858	Welding: Aluminum, Magnesium, Non-hardening steel Or Alloys, and Titanium, Spot, Seam and Switch
MIL-S-6872	Soldering Process, General Specification For
MIL-L-7870	Lubricating Oil, General Purpose, Low Temperature
MIL-I-8660	Insulating and Sealing Compound, Electrical
MIL-T-9107	Test Reports; Preparation Of
MIL-L-10547	Liners, Case, Waterproof

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MS33586	Metals, Definition of Dissimilar

Air Force-Navy Aeronautical

AN3102	Connectors - Electrical, Receptacles, Box Mounting
AN3106	Connectors - Electrical, Plugs, Straight

DRAWINGS

U. S. Air Force

56J22785	Rack Assembly - Bomb, Electrically Operated, 5000 Pound Capacity, Type MB-3A

PUBLICATIONS

Air Force-Navy Aeronautical Bulletin

143	Specifications and Standards; Use Of

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Preproduction.- This specification makes provisions for preproduction testing.

3.2 Selection of specifications and standards.- Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with ANA Bulletin 143.

3.3 Materials

3.3.1 Protective treatment.- When materials are used in the construction of the rack that are subject to corrosion in salt air or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in accordance with QQ-P-416. Acceptance or approval of material in the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product.

3.3.2 Metals.- Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.3.3 Dissimilar metals.- Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.4 Design and construction.- The design and construction of the bomb rack shall conform to Drawing 56J22785 and this specification.

3.4.1 Soldering.- Soldering required in the fabrication of the rack shall be in accordance with MIL-S-6872.

3.4.2 Welding.- Any welding required in the fabrication of the rack shall be in accordance with MIL-W-6858.

3.4.3 Springs.- Care shall be exercised in the treatment of springs to prevent methods of manufacture which will result in brittleness. To prevent brittleness as a result of plating, all springs shall be treated in accordance with the applicable detail drawing.

3.5 Performance

3.5.1 Shock.- The bomb rack shall not function inadvertently when subjected to shock tests specified in 4.6, Test Methods.

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3.5.2 Vibration.- The bomb rack shall be capable of satisfactory operation in accordance with 3.6 during and after the vibration tests described in 4.6.12.

3.5.3 Temperature.- The bomb rack shall be capable of satisfactory operation in accordance with 3.6 throughout the temperature range of -65°F to + 160°F.

3.5.4 Lubrication.- The bomb rack shall operate satisfactorily without lubrication. If an initial lubrication is necessary in the sealed units, lubricants shall conform to MIL-G-3278 or MIL-L-7870.

3.5.5 Manual operation.- Means shall be provided for easy cocking and releasing of the rack with or without a store installed.

3.5.5.1 Cocking force.- A force of no more than 40 pounds shall be required to hand cock the rack.

3.5.5.2 Tripping force.- A force of no more than 7 pounds shall be required to trip the rack when a total load of 40,000 pounds is applied.

3.5.6 Arming mechanism.- When the bomb rack is cocked, the arming wire retainers shall allow the arming wire swivel loops to be inserted and held in place. When the rack is released with the arming solenoid energized, the arming wires shall be retained by the solenoid plungers. Upon de-energization of the solenoid (see figure 1), the arming wires shall be released with any dead load up to and including 3 pounds. No load shall be required to release the wires when the rack is released with the solenoids un-energized. Rack release with the solenoids energized is known as "armed" release. Release with the solenoids un-energized is known as "safe" release.

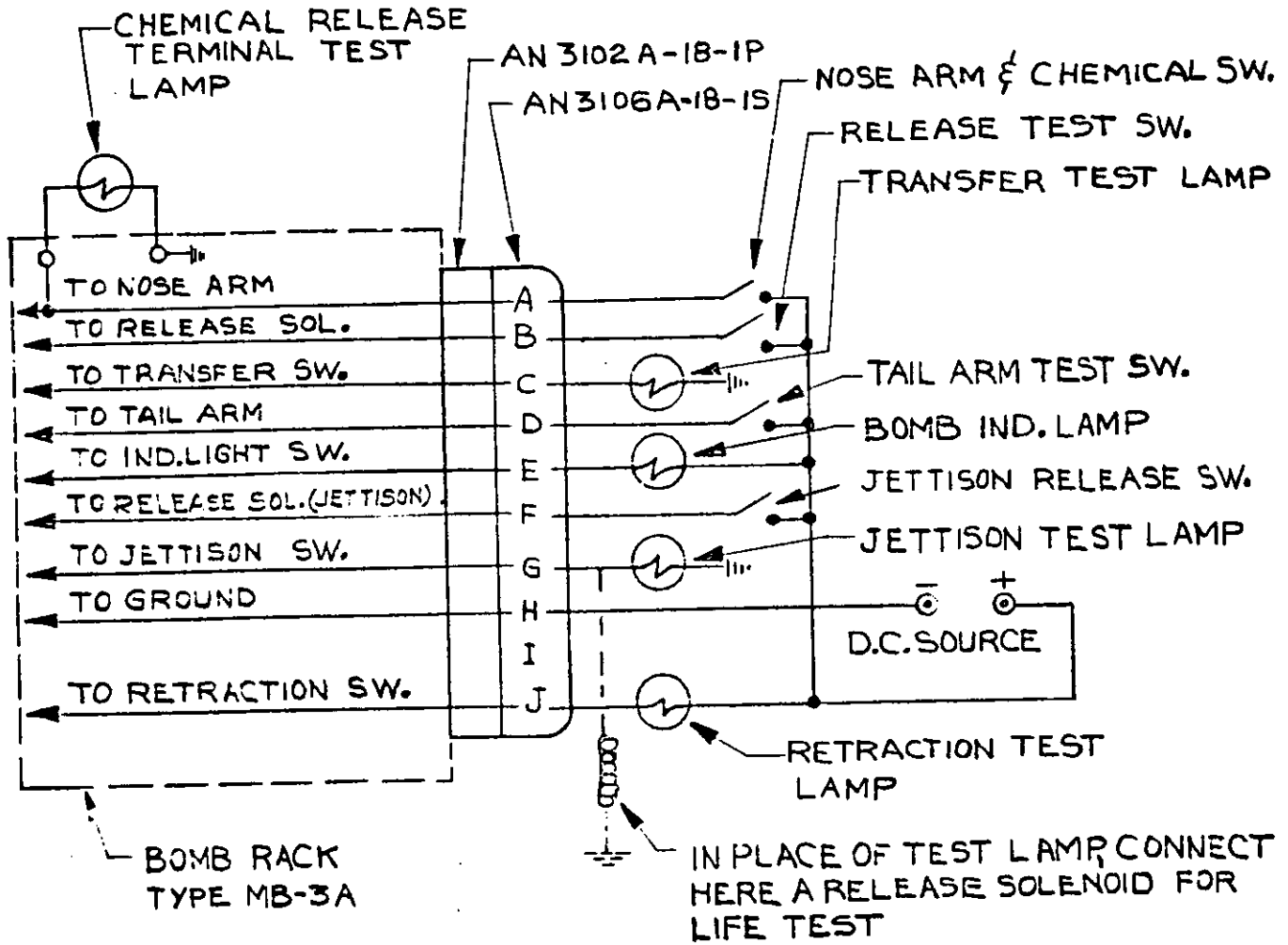
3.6 Electrical characteristics

3.6.1 Voltage.- The bomb rack shall fulfill all the operating requirements as specified herein when supplied any voltage from 18V to 28V dc.

3.6.2 Current

3.6.2.1 The current drain of the release solenoid shall not exceed 6 amperes at 24V dc, at a room temperature of approximately 58°F (20°C). The current drain shall be measured with the solenoid installed in the bomb rack.

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ALL LAMPS 24 VOLTS

FIGURE 1

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3.6.2.2 The current drain of the arming solenoids shall not exceed 0.55 ampere at 24V dc, at a room temperature of approximately 68°F (20°C). The current drain shall be measured after the solenoid has been energized for 2 minutes and while it is installed in the bomb rack.

3.6.3 Time of release.- Not more than 0.030 second shall elapse from beginning of electrical impulse through release of circuit "B" in figure 1 to the instant when bomb carrying hooks begin to open.

3.6.4 Time of transfer.- During operations of release circuit "B" in figure 1, the time from the end of the electrical impulse to the completion of transfer into the next rack release circuit shall be a maximum of 0.030 second.

3.6.5 Release impulse.- The release mechanism shall operate using a release impulse duration of 0.020 second or less. With a release impulse of 0.020 second, the combined time of release and transfer shall not exceed 0.075 second. The above time tests shall be conducted using 18V dc, supplied to the rack and a room temperature of approximately 68°F (20°C). The combined time of release and transfer shall be measured from the beginning of the release pulse until a pulse indicating the completion of transfer is obtained through the "C" terminal of the bomb rack.

3.6.6 Jettison release circuit.- There shall be a separate release circuit (jettison) in paralled to the normal release circuit, by means of which the rack can be released independently of the normal release circuit. This circuit shall be such that, immediately upon opening of the hooks, the impulse required to release the rack is transferred to a jettison release output wire. With this circuit, it shall then be possible to release several racks with one impulse.

3.7 Interchangeability.- All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-5028 shall govern changes in the manufacturer's part numbers.

3.8 Weight.- The weight of the rack shall not exceed 22 pounds.

3.9 Identification of product.- Equipment, assemblies and parts shall be marked for identification in accordance with MIL-STD-130.

3.10 Workmanship.- The bomb racks shall be uniform in quality and material, of high quality workmanship, and free of any injuricus defects.

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3.10.1 Cleaning.- Each rack shall be thoroughly cleaned, and the mechanism and rack housing shall be entirely free of loose particles such as dirt, metal chips, loose bits of wire and solder, etc.

3.10.2 Sand blasting.- All parts which are sand blasted shall be thoroughly cleaned to remove sand and grit.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Classification of tests.- The inspection and testing of the MB-3A bomb rack shall be classified as follows:

a. Preproduction tests

b. Acceptance tests

4.3 Preproduction testing

4.3.1 Preproduction test sample.- When so specified in the contract a minimum of 4 bomb racks shall be tested at a laboratory designated by the procuring activity, or, when so specified in the contract, at the contractor's plant under the supervision of the procuring activity. It is not required that all tests be conducted on any one test sample.

4.3.2 Preproduction tests.- Preproduction tests shall consist of all tests described under 4.6, Test Methods.

4.3.2.1 Racks subjected to tests described under 4.6.10.1 shall be treated as follows. The entire internal linkage of the rack shall be sprayed with a liquid mixture consisting of 8 ounces of Dow Corning DC-4 or DC-4A Silicon Compound, which conforms to MIL-I-8660, and 1 pint of methylene chloride. After being sprayed, the rack shall be permitted to dry for 24 hours prior to conducting these tests.

4.4 Acceptance tests.- Acceptance tests shall consist of:

a. Individual tests

b. Periodic sampling tests

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4.4.1 Individual tests.- Each bomb rack shall be subjected to the following tests as described under 4.6, Test Methods, in the same order:

a. Examanation of product

b. Functional tests

c. Shock test

d. Proof test

4.4.2 Periodic sampling tests.- Racks to be subjected to "Sampling Tests" shall be selected at random from production units which have successfully completed all individual tests. Sampling tests shall consist of all tests described under "Test Methods", excluding 4.6.10.1.2, 4.6.10.2, 4.6.10.3, 4.6.10.4 and 4.6.11. Racks shall be selected in the following manner:

a. Not less than 1 nor more than 4 racks from the first 100 production units on the contract or order.

b. Not less than 1 nor more than 4 racks from each succeeding 250 units or fraction thereof on the contract or order.

4.4.2.1 Rejection and retest.- When one rack fails to meet the specification, acceptance of all racks on hand will be withheld until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.4.2.2 Individual tests may continue.- For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of the racks on hand or produced later shall not be made until it is determined that the racks meet all the requirements of the specification.

4.4.2.3 Periodic test samples.- Racks shall be subjected to tests in such a manner as to conserve sample racks. It is not required that all tests be conducted on any one rack.

4.5 Test conditions.- Unless otherwise specified, tests required by this specification shall be made under normal ambient conditions at approximately 68°F.

4.6 Test methods

4.6.1 Examination of product.- Each rack shall be inspected for conformance to the drawing and specification with respect to material, workmanship, dimensions, and markings.

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4.6.1.1 Workmanship inspection.- Each part of assembly shall be inspected for the existence of any of the conditions specified below. The occurence of any one of the following conditions shall be sufficient reason for rejection:

a. Dirty or pitted electrical contact points
b. Dirt, metal, chips, filings, solder or burrs on any surface
c. Visible lubrication
d. Parts not treated for corrosion resistance
e. Loose connections, badly soldered joints or frayed ends of wires
f. Possible short circuiting because of proximity of bare wires or switch parts to metal portions of the rack
g. Play in levers and latches which may permit accidental release because of side slip or misalignment
h. Loose rivets, screws, or pins
i. Improper matching of contact points on electrical switches in both the cocked and released position
j. Misalignment of parts
k. Insufficient clearance between moving parts and the case
l. Clearance of openings through which dust may enter vital parts
m. Presence of moisture or liquids

4.6.1.2 Detailed inspection of critical items.- Extensive experimental tests have proved that if any one of the following unsatisfactory conditons exist, failure of the bomb rack will result after a short period of usage:

a. Each bomb rack shall be inspected to insure that the wire cable leading from the receptacle into the solenoid and switch housing can in no way, under any condition, interfere with the operation of the arming device.

b. Each bomb rack shall be carefully inspected to insure that the 0.406 $\pm$ 0.001 inch hole in the brass bushing of the solenoid and switch housing assembly through which the release lever rotates in concentric with the 0.281 $\pm$ 0.001 inch hole in the frame assembly, as specified on the listed drawings.

c. Each bomb rack shall be inspected to insure that hooks shall rotate freely after mounting screws have been secured. The inspection shall include the critical dimensions of the associated parts such as the dimensions of the length of the shoulders of the pivot bushings, the dimensions across the faces of the hook bosses, and the size of the frame hole.

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d. Release lever locking arms shall be inspected to insure that the angle and the dimensions which locate the 1/8 inch land with respect to the center of rotation are within the specified limits.

e. An inspection shall be made to insure that binding will not exist between the pivot hole of the cocking lever and the release spring bushing after a short period of operation.

4.6.1.3 Bomb rack hook hardness test.- All hooks shall be hardness tested for conformance to the drawings.

4.6.2 Functional tests

4.6.2.1 Arming mechanism functioning.- The arming mechanism shall be checked for proper function in accordance with 3.5.6.

4.6.2.2 Electrical circuits.- One terminal of each solenoid is grounded to the solenoid case, which is in turn grounded to the bomb rack housing. Figure 1 of this specification shows the wiring diagram governing the electrical connections necessary to the operation of this bomb rack. The electrical system for the rack shall be tested for proper operation as follows:

a. Cock rack and attach electrical connector. Bomb indicator lamp shall light.

b. Operate release test switch. Rack hooks shall snap open and bomb indicator light turn off. Retraction circuit test light shall light.

c. Operate release test switch again, and transfer circuit test lamp shall light while switch is held "On".

d. Operate nose arming solenoid and chemical switch. Nose arming solenoid plunger shall extend into slot of sliding member and chemical release terminal test lamp should light while switch is in the "on" position.

e. Operate tail arming switch. Tail arming solenoid plunger shall extend into slot of sliding member.

f. Both arming solenoids shall satisfactorily suspend and release a 3 pound load at the option of the operator.

g. Cock rack. Operate jettison release switch. Bomb rack hooks shall snap open and jettison test lamp shall light as soon as hooks open.

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4.6.2.3 Each release solenoid shall be energized for a period of 30 seconds at 24V dc without resulting damage.

4.6.2.4 Current drain test.- Each bomb rack solenoid shall have current drain checked to determine conformance to 3.6.2.1 and 3.6.2.2.

4.6.2.5 Time of release tests.- Measurements shall be made to determine the length of time necessary from beginning of electrical impulse to the instant when the bomb carrying hooks begin to open. Not more than .030 second shall elapse. Measurements for this test and for that described in 3.6.3 are to be made with 18V dc applied to the rack.

4.6.2.6 Time of transfer tests.- Tests shall be accomplished with 18V dc applied to the solenoid to determine time of transfer for conformance to 3.6.4.

4.6.2.7 Release impulse.- Tests shall be conducted to determine compliance with conditions set forth in 3.6.5 which govern the release impulse.

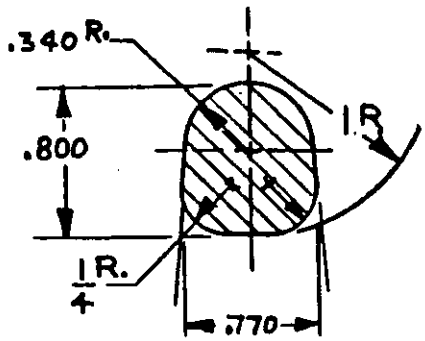
4.6.3 Shock test.- Each rack shall be subjected to a shock test. The method of shock testing employed by the contractor shall be subject to the approval of the procuring activity. This test is to determine whether the release solenoid will rotate under shock load permitting the rack linkage to release. This test shall be conducted with no load on the bomb rack hooks. The shock load may be applied by use of a swinging pendulum or a jig for dropping the rack. The shock load shall be a minimum of 15g, for a 12 millisecond time duration, applied near the release solenoid and acting in such a direction as to tend to cause the solenoid to release the rack linkage. Bomb racks shall be rejected which are triggered during the shock test.

4.6.4 Proof test.- Each rack shall be proof loaded after complete assembly by applying a load of 25,000 pounds suspended from the carrying hooks as shown in figure 2. The position of the load and the cross section of the pin engaging the carrying hooks shall be in accordance with the dimensions specified in figure 2. After electrical release of the rack, all parts shall function freely. Upon completion of the proof test, tests in 4.6.2 shall be repeated.

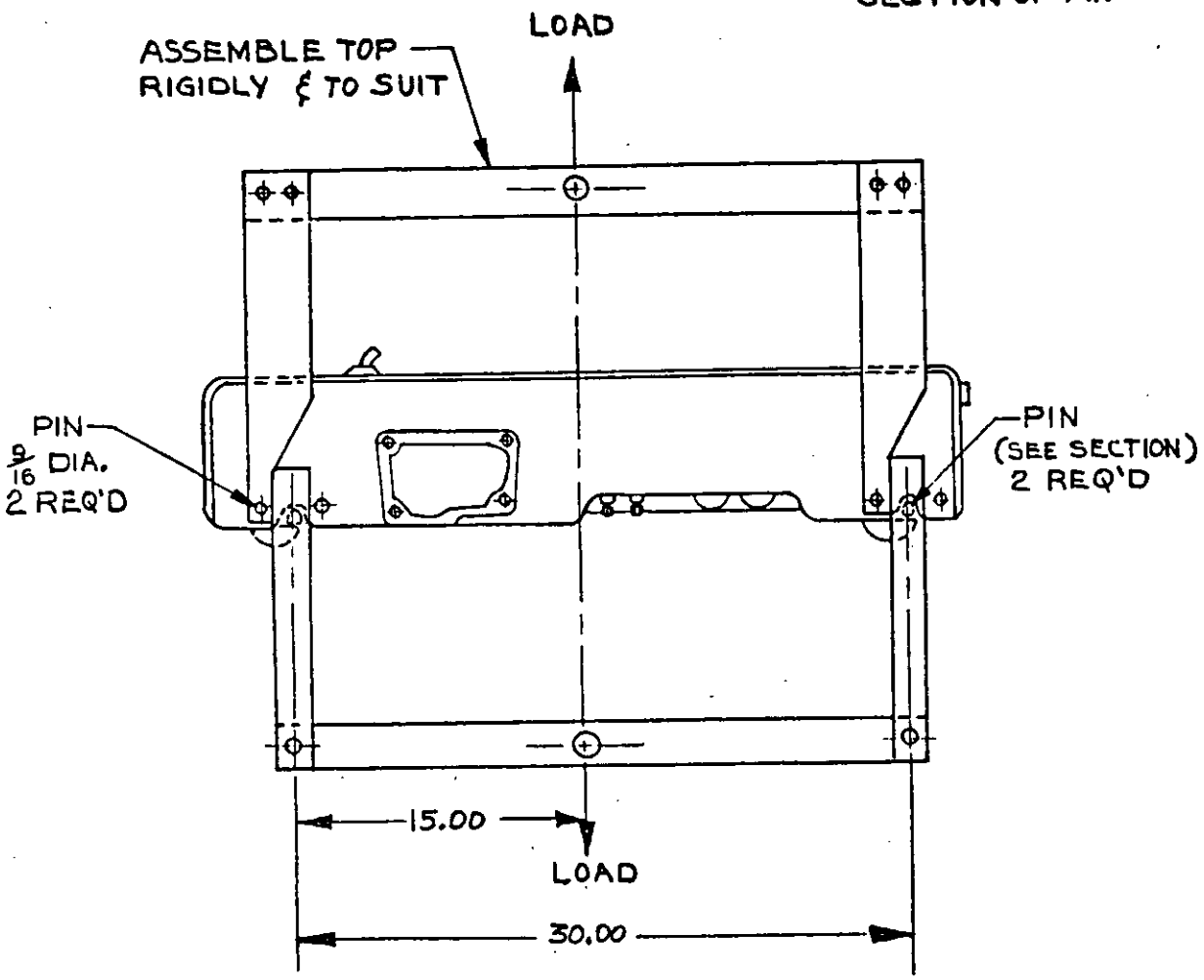
4.6.5 Bomb carrying hooks

a. Heat treatment.- A check of the heat treatment of the hooks shall be made by means of not less than three test bars heat treated with each group of hooks. The tensile strength designated on the hook drawings is the minimum allowable and the elongation shall not be less than that specified in the material specification for the grade of steel given on the drawing.

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SECTION OF PIN



LOADS:

PROOF LOAD	25,000 LB.
MAX. ELECTRIC RELEASE LOAD	40,000 LB.
ULTIMATE LOAD	50,000 LB.

DIMENSIONS IN INCHES

FIGURE 2

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b. Ultimate strength. - Two hooks shall be tested to destruction from each group heat treated. The hooks shall be tested in a fixture which simulates their actual assembled position in a cocked bomb rack. Each hook shall withstand a minimum load of 45,000 pounds.

4.6.6 Cocking force test. - The cocking mechanism shall be tested to determine that a force of no more than 40 pounds is required to hand cock the rack.

4.6.7 Explosion proof test. - The rack shall be subjected to the explosion proof test in accordance with MIL-E-5272, procedure I. One explosion during the test shall indicate a failure to meet the requirements.

4.6.8 Arming solenoid tests. - These tests shall be conducted with the solenoids installed in the bomb rack.

a. Both the arming solenoid plungers shall be loaded with a downward vertical pull (see figure 3) of 101 pounds when locked in the "armed" position by energizing their respective solenoids. After removal of the above specified load, the solenoid shall be de-energized. After this test the solenoid shall comply with 3.6.2.2 and 4.6.2.2.f. There shall be no visual deformation or bending of the solenoid plunger that will prevent its satisfactory operation in accordance with this specification.

b. The arming solenoids shall be life tested at 24V dc by energizing the solenoid for 24 hours. No damage shall result from this continuous operation.

4.6.9 Release solenoid life test. - The release solenoid, while installed in the bomb rack, shall be energized by 24V dc for a minimum of 3 minutes. At the end of this test, the solenoid must be operative, although damage to insulation and changes in current and voltage characteristics may have resulted.

4.6.10 Environmental tests. - There shall be no mechanical or electrical failure as a result of these tests.

4.6.10.1 High and low temperature release and transfer tests.

4.6.10.1.1 The bomb rack shall be loaded with not more than 50 pounds per hook (100 pounds total) and subjected to a temperature of -65°F or colder for a period of 24 hours. The bomb rack shall then be tested at this temperature with 18V dc supplied to the rack. The time of release shall be in accordance with 3.6.3. The time of transfer shall be as specified in 3.6.4. The arming mechanism shall function satisfactorily as specified in 3.5.6.

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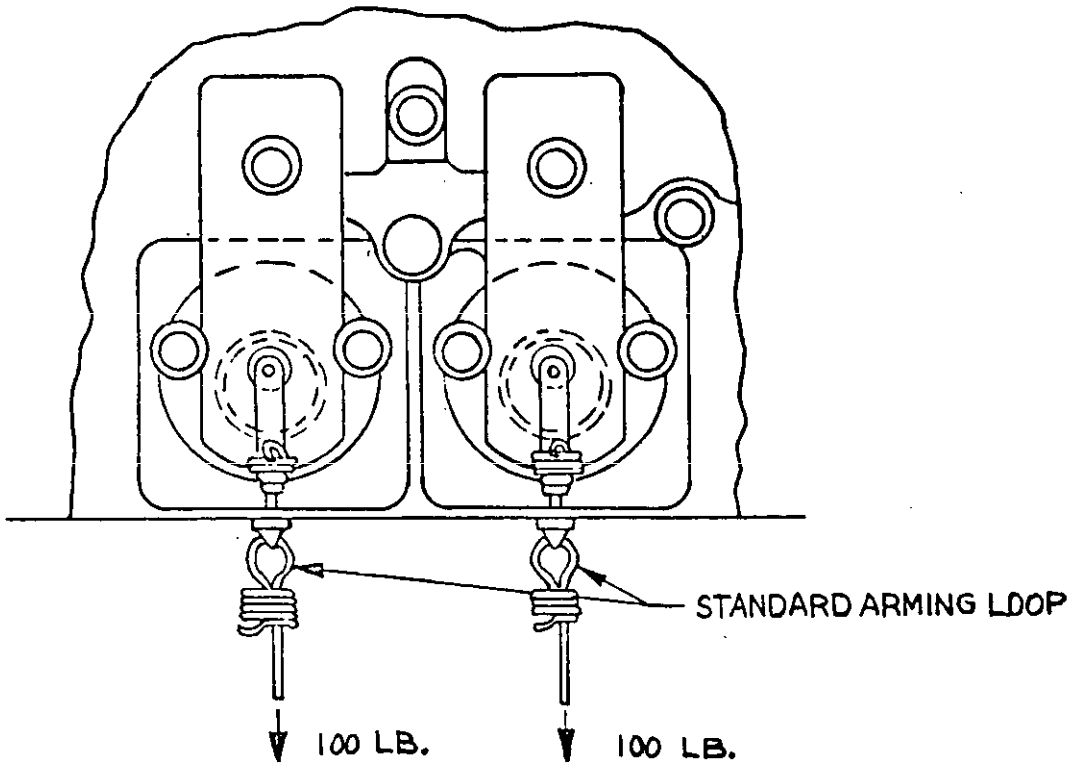


FIGURE 3

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4.6.10.1.2 After being tested as specified in 4.6.10.1.1, the rack shall be exposed to an atmosphere of $\pm 76^{\circ}F$ and a relative humidity of 98 percent until moisture has collected over the entire rack. The rack shall then be loaded with a maximum of 50 pounds per hook and subjected to a temperature of $-65^{\circ}F$ or colder for 5 hours and operated. The time of release or time of transfer need not be checked; however, the rack shall release without apparent hesitation and the arming wire retainers and arming solenoids shall function properly.

4.6.10.1.3 The rack shall be subjected to a temperature of $160^{\circ}F$ for a period of 24 hours, and the time of release and the time of transfer checked. The time of release and time of transfer each shall not exceed 0.030 second using 18V dc. A check shall be made to determine that this temperature does not adversely affect any of the component parts.

4.6.10.2 Sand and dust test.- The rack shall be subjected to the sand and dust test in accordance with MIL-E-5272, procedure II. At the conclusion of this test, the rack shall release without hesitation with no load applied to the hooks.

4.6.10.3 Humidity test.- The rack shall be subjected to the humidity test in accordance with MIL-E-5272, procedure I. At the conclusion of this test, the rack shall release with no load applied, after being thoroughly dried.

4.6.10.4 Salt spray test.- The rack shall operate properly after being subjected to the salt spray test in accordance with MIL-E-5272, procedure I.

4.6.11 Limit load test.- The operation of the rack shall not be adversely affected by the suspension of a load of 33,000 pounds for a period of not less than 3 minutes. There shall be no visually noticeable deformation of any of the components or assemblies as a result of this test.

4.6.12 Vibration tests

4.6.12.1 The bomb rack shall be cocked and mounted on a vibration stand with a line joining the centers of the mounting bolts in a horizontal plane and with the plane of the bosses of the bomb rack vertical. The bomb rack shall be subjected to vibration with a single amplitude between 0.0015 and 0.0025 inch (total excursion of 0.003 to 0,005 inch) at frequencies of 500 to 3000 cycles per minute for a sufficient length of time to determine that the natural frequency of the bomb rack does not occur in this frequency range.

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4.6.12.2 After 4.6.12.1, the rack shall be vibrated for a 3-hour period at various frequencies between 1000 and 2500 cycles per minute, with an amplitude of from 0.009 to 0.010 inch (total excursion of 0.018 to 0.020 inch). During this test, the rack shall not trip. During and after this test, the electrical characteristics shall conform to the requirements of 3.6. No looseness in the mechanism or damage to any part of the bomb rack shall result from the 3 hour vibration period.

4.6.13 Life test. - The bomb rack shall be electrically released with 28,000 pounds suspended from the hooks through a total of 1000 cycles at the rate of not less than 60 cycles per hour. Each cycle shall consist of cocking the unit, applying the load, and electrically releasing the rack. At the end of 1000 cycles of operation, no parts shall be broken nor shall there be any damage or wear sufficient to affect adversely the operation of the rack. The first half of the test shall be accomplished using the normal release circuit "B". The remaining portion of the test shall be accomplished using the jettison circuit with sufficient time between cycles to energize a release solenoid connected to the salvo output lead of the connector (see figure 1). At the end of this test, the rack shall be checked for a release through the normal release circuit. This test shall be conducted using 24V dc.

4.6.14 Tripping force.- The force required to trip the rack shall be measured to determine conformance to 3.5.5.2.

4.6.15 Maximum electrical release load.- A bomb rack shall be loaded to 40,000 pounds. The load shall be released electrically with 18V dc applied. The rack must release the load although damage may result.

4.6.16 Ultimate strength test.- A rack shall be selected for the ultimate strength test and shall be proof loaded to 25,000 pounds in accordance with figure 2. The load shall be released electrically. The bomb rack shall then be loaded at a uniform rate to 50,000 pounds ultimate load within 3 minutes.

5. PREPARATION FOR DELIVERY

5.1 Levels of preservation and packaging

5.1.1 Level A.- The rack shall be preserved in accordance with MIL-P-116, method IIB and individually packaged in containers meeting PPP-B-636.

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5.1.2 Level C.- The rack shall be preserved and packaged in accordance with manufacturer's commercial practice.

5.2 Levels of packing

5.2.1 Level A.- The rack shall be packed in exterior type shipping containers meeting PPP-B-591, PPP-B-621, PPP-B-585, PPP-B-636, MIL-B-138, or PPP-B-601. As far as practical, exterior containers shall be of uniform shape and size, be of minimum cube and tare consistent with the protection required, and contain identical quantities. The gross weight of each pack shall be limited to approximately 200 pounds. Containers shall be closed and strapped in accordance with the applicable container specification. Containers shall be provided with a case liner conforming to MIL-L-10547 and sealed in accordance with the appendix thereto. The case liner will not be required when the unit, intermediate, or exterior container conforms to PPP-B-636 and is sealed at all joints and seams with tape conforming to PPP-T-60.

5.2.2 Level C.- Packages which require over-packing for acceptance by the carrier shall be packed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall meet Consolidated Freight Classification rules or regulations of other common carriers as applicable to the mode of transaportation.

5.3 Marking.- Interior and exterior shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

Rack, Bomb, Aircraft, Electrically Operated, Type MB-3A
Specification MIL-R-25085C(USAF)

6. NOTES

6.1 Intended use.- This bomb rack is intended for use in fighter and light, medium, and heavy bombardment aircraft to suspend, release, and arm bombs up to and including 5000 pounds with provisions for carrying and releasing chemical munitions.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable levels of preservation and packaging and packing.

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6.3 The type MB-3A bomb rack is intended for use on airplanes carrying bombs containing high explosives; it is therefore necessary to obtain units which are obsoultely free from defects in workmanship and assembly in order that the hazards resulting from the use of these explosives are minimized. Therefore, it is stressed that every effort shall be exercised to obtain racks which are free from defects in workmanship, material and assembly.

6.4 The use of jigs, fixtures, and dies is recommended for use in the construction of bomb racks of this type to obtain complete units having uniform dimensions, thereby permitting interchangeability of parts.

6.5 Definitions

6.5.1 Ultimate strength.- Maximum load the unit shall be capable of withstanding for a period of not less than 10 seconds.

6.5.2 Maximum release load.- Maximum load at which unit can be released although damage may result.

6.5.3 Limit load.- Maximum applied load at which no damage or deformation occurs to assembly or any components.

6.5.4 Proof load.- Load applied to each production unit after final assembly.

6.5.5 Operating release load.- Load at which unit can be repeatedly released without damage occuring.

6.5.6 Rated load.- Maximum static weight of store which unit is designed to carry.

6.5.7 Maximum tripping force.- Force required to manually trip a unit with a specified load applied.

NOTICE: When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

PREPARING ACTIVITY
APGC

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEM PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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